SAKINA SAIFUDDIN

sakinasaifee006@gmail.com . Houston,Texas . +1 (346) 870 0136 . LinkedIn . Portfolio . Github

SUMMARY

MS Data Science & AI candidate with 3.5+ years shipping production AI/ML, data pipelines, and BI systems across banking, manufacturing, and healthcare. Automated 1,074+ manual reports (50% workload cut), lifted medical-imaging classification to 91%, and boosted CNN accuracy to 86.65% via GAN augmentation. Fluent in Python, SQL, LLMs, ETL, and Power BI.

EDUCATION

Master of Science in Engineering Data Science & AI, University of Houston, Houston,TX. Aug 2025 – Present
Coursework: Predictive Data Analytics · Machine Learning · Applied Neural Networks · Big Data Analytics · Data Visualization · Data Science · Database Management · Software Engineering
Bachelors in Computer Science (Gold Medalist), Iqra University, Pakistan, GPA: 3.89/4.0 Jan 2021 – Jan 2025

PROFESSIONAL EXPERIENCE

Dental-X AI – Los Angeles, CA May 2026 – Present
AI Automation Engineer (Intern)
- Architected a Python ETL pipeline unifying 15–20 third-party APIs (CareStack, DentalXchange, SQL sources) into a single canonical schema, replacing manual lookups and saving ~2 hrs per case.
- Lifted data accuracy from 20% to 80% by engineering a normalization layer that reconciles conflicting payer responses into standardized codes, validated in BigQuery across 7 production deployments.

Texas Manufacturing Assistance Center (TMAC / NIST-MEP) - Houston, TX Feb 2026 – Present
Business Intelligence & Automation Intern
- Automated 1,074+ manual Excel reports across 11+ departments and 50+ grants with Python AI pipelines over PeopleSoft/SQL, cutting workload 50% within the first month of deployment.
- Built LLM-assisted reconciliation via REST API that resolved general-ledger discrepancies 3× faster with zero engineering hand-offs to finance leadership.
- Delivered 10+ Power BI / PowerApps dashboards converting 50+ grants of raw data into one-click financial insights for senior leadership.

University of Houston — Department of Mathematics - Houston, TX
Graduate Research Assistant Dec 2025 – Feb 2026
- Trained SVM & CNN (Keras) models to 91% classification accuracy on 1,000+ neuron microscopy images, quantifying GSK3-inhibitor drug impact under Prof. Labate Demetrio.
- Engineered a Python/OpenCV/Scikit-learn pipeline that automated image extraction, cutting manual analysis time and enabling reproducible, at-scale research workflows.

BluTech Consulting - Karachi, Pakistan May 2024 – Aug 2025
Business Intelligence Consultant
- Delivered 10+ Power BI & Tableau dashboards over 500K+ records via SQL, Teradata & HiveQL for UBL (Pakistan's top-5 bank), surfacing real-time performance bottlenecks.
- Cut reporting turnaround 25% by automating anomaly-detection pipelines and delivering root-cause findings to non-technical executive leadership.

SoftSyncDev - Karachi, Pakistan Jan 2021 – May 2024
Software Engineer
- Shipped a production AI chatbot via OpenAI REST API serving 1,000+ users; resolved Prisma/Node.js bottlenecks to cut response latency 40%. Demo
- Deployed 5+ full-stack SaaS apps on the MERN stack with AWS S3 & CI/CD pipelines, sustaining 99.9% uptime. Demo

PROJECTS

GAN-Based Chest X-ray Augmentation for Pneumonia Detection | GitHub
- Built a DCGAN pipeline (Torchvision) generating 2,000 synthetic images to fix class imbalance on a ~5.2K Chest X-ray dataset; retrained CNN on the 9.2K-image set, lifting test accuracy from 73.56% to 82.21%.

AI-Driven Retail Inventory Forecasting & Autonomous Inventory Optimization | GitHub
- Built an end-to-end demand forecasting pipeline (Linear Regression, Random Forest, LightGBM) on 3M+ sales records.
- Engineered time-series features with K-Fold validation as the predictive layer for a Deep Q-Network autonomous replenishment system.

HMS — Harmful Brain Activity Classification | GitHub
- Trained CNN & EfficientNetB2 models to classify harmful brain activity (seizure vs. non-seizure) from 54,000 EEG spectrograms, evaluated via Kullback–Leibler divergence.

Handwritten Digit Recognition | GitHub
- Developed a deep learning model achieving 98.7% accuracy on MNIST and 95% on unseen images as part of Applied Neural Networks coursework.

SKILLS

Languages & Tools: Python (OOP), SQL (Hive, Teradata, Oracle), JavaScript, TypeScript, Java, Node.js, AWS (S3), Git, CI/CD
ML / AI: OpenAI API, LLM Integration, Prompt Engineering, TensorFlow/Keras, PyTorch, Scikit-learn, CNNs, GANs, SVMs, LightGBM, DQN, Time-Series Forecasting, NLP, Hugging Face, Model Versioning
Data & Analytics: Pandas, NumPy, Random Forest, Anomaly Detection, Predictive Analytics, K-Fold Validation, OpenCV, EDA **BI & Visualization:** Power BI, Tableau, PowerApps, SSRS, DBeaver, Matplotlib, Seaborn, Looker (familiar), Snowflake (familiar) **Web / App Dev:** React.js, React Native, Next.js, Vue.js, Express.js, MongoDB, MERN Stack, Prisma, REST APIs, Redux Tool